

Homework 5

ORF 245 - Fundamentals of Statistics

Due: Mar. 28th, 2025, End-of-Day (11:59pm)

Notes:

- Collaboration with other students is encouraged, but you must i) write up your own answers, and ii) acknowledge any sources or students you consulted.
 - Submit the homework on Gradescope, and be sure to assign each page to the relevant question(s).
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Exercise: 1. (Law of Large Numbers)

Let $X_1, \dots, X_n \stackrel{iid}{\sim} \text{Poisson}(2)$.

- (a) Define new random variables $Y_1, \dots, Y_n$, such that

$$Y_i = \begin{cases} 1 & \text{if } X_i = 2 \\ 0 & \text{otherwise.} \end{cases}$$

What is the joint distribution of $Y_1, \dots, Y_n$?

(Note: Another way we could define Y_i is: $Y_i = 1\{X_i = 2\}$. The function “ $1\{\cdot\}$ ” equals one if the condition inside the brackets is true, and zero otherwise. You will encounter this notation later in this problem.)

- (b) As $n \rightarrow \infty$, what value does $\bar{Y}_n := \frac{1}{n} \sum_{i=1}^n Y_i$ approach? Why?
- (c) More generally, if $Z_i = 1\{X_i \in [a, b]\}$, for some two values $a < b$, what value will $\bar{Z}_n := \frac{1}{n} \sum_{i=1}^n Z_i$ approach as $n \rightarrow \infty$?
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Exercise: 2. (Simulation: Law of Large Numbers)

Let $X_1, \dots, X_n \stackrel{iid}{\sim} \text{Exp}(\frac{1}{2})$. The marginal pdf of each X_i is hence given by:

$$f_X(x) = \begin{cases} \frac{1}{2}e^{-\frac{x}{2}}, & 0 \leq x < \infty \\ 0, & \text{else} \end{cases} \quad (1)$$

$\bar{X}_n = \frac{1}{n} \sum_{i=1}^n X_i$ denotes the random sample mean as usual.

- (a) For what values of n will we have $\text{Var}(\bar{X}_n) \leq 0.01$?
- (b) Perform the following simulation in R – for every n between $n = 1$ and $n = 2000$:
- (i) Generate realizations $x_1, \dots, x_n$ of the joint random variables $X_1, \dots, X_n$.
(Recall these are assumed to be i.i.d. from an Exponential distribution with parameter $\frac{1}{2}$.)
 - (ii) Compute the corresponding realization of the sample mean $\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i$ and store it.

Plot the following two lines on the same figure. What do you observe and why?

- The realizations of the sample mean $\bar{x}_n$ vs the sample size n .
- The expected value of the random sample mean $\mathbb{E}[\bar{X}_n]$ vs n . (*Hint: This should be a horizontal line.*)

To do this exercise you may need to look up: 1) how to do for loops in R; 2) how to generate random samples from an exponential distribution.

Exercise: 3. (Simulation: Central Limit Theorem)

Let $X_1, \dots, X_n \stackrel{iid}{\sim} \text{Exp}(\frac{1}{2})$, and $\bar{X}_n := \frac{1}{n} \sum_{i=1}^n X_i$ denotes the random sample mean as usual.

- (a) How many samples n are needed to guarantee that the variance of $\bar{X}_n$ is no larger than 0.01?
- (b) Let $n = 1000$. In R, generate realizations $x_1, \dots, x_{1000}$ of the i.i.d. random variables $X_1, \dots, X_{1000}$. Plot the histogram (of the values $x_1, \dots, x_{1000}$), with bins of any size ensuring the resulting figure is informative.

What does this histogram look like? Why?

- (c) Let $n = 1000$. We can obtain a single realization of the random sample mean $\bar{X}_{1000}$ by computing $\bar{x}_{1000} = \frac{1}{1000} \sum_{i=1}^{1000} x_i$, where $x_1, \dots, x_{1000}$ are realizations of the joint random variables $X_1, \dots, X_{1000}$.

Generate 5000 realizations of $\bar{X}_{1000}$ and plot the histogram of these 5000 values. Again, you may choose bins however you like as long as the resulting figure remains informative.

What does this histogram look like? Why? What is the approximate distribution of $\bar{X}_{1000}$?

Exercise: 4. (Central Limit Theorem)

Let $X_1, \dots, X_{100} \stackrel{iid}{\sim} \text{Ber}(0.2)$, and define as usual $\bar{X}_{100} := \frac{1}{100} \sum_{i=1}^{100} X_i$.

- (a) Compute $\mathbb{P}(\bar{X}_{100} > 0.22)$, in terms of the standard normal CDF $\Phi(\cdot)$.

- (b) Compute $\mathbb{P}(|\bar{X}_{100} - 0.2| > 0.02)$, in terms of the standard normal CDF $\Phi(\cdot)$.
- (c) Compute $\mathbb{P}(\bar{X}_{100} > 0.22)$ **without** using the standard normal CDF. How different is your answer from that of part (a)? Why?
- (Hint: What is the distribution of $\sum_{i=1}^{100} X_i$, if $X_1, \dots, X_{100}$ are iid Bernoulli random variables?)